

1. It converges to the value of the start state $v_{\pi}(S)$ (or $v_{\pi_{\epsilon}}(S)$ if you're using the ϵ -greedy policy), since every rollout begins from S . The sample mean of discounted returns across episodes converges to this expectation by the Strong Law of Large Numbers, because each episode's return is an independent and identically distributed sample with mean $v_{\pi}(S)$.
2. Policy $\pi_{0.2}$ results in lower returns because $\pi_{0.2}$ is more exploratory, it makes more suboptimal moves. When traversing the safe row above the cliff, a "wrong" action of Down drops you into the cliff for -100 and a reset to S . With $\epsilon=0.2$, the chance of picking Down on any step is $0.2 / 3 \approx 0.067$. Over ~ 10 right-steps above the cliff, the chance of *avoiding* a drop is $(1-0.067)^{10} \approx 0.50$, about a 50% chance to fall at least once which lowers the average return. Even when it doesn't fall, extra random moves add detours, piling up -1 per step and delaying the goal. So the higher ϵ policy suffers more frequent large penalties and greater cumulative step costs, yielding much lower returns.